FrostBit DEX Technical Whitepaper

Snowman Consensus, Cross-Chain Custody, and On-Chain Trading

CommonstateLab

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Implementation baseline FrostBit v0.8.4

Abstract

FrostBit is an application-specific blockchain for spot and perpetual trading. Snowman orders and finalizes blocks through repeated sampling. Its block execution engine settles trades and audits token conservation. Cross-chain vaults use FROST threshold signatures and an independent Guardian authorization. Deposits are credited after evidence is evaluated by witnesses and exposed to challenge and arbitration. This revision describes the rules in the v0.8.4 source, the assumptions behind those rules, and the measurements needed to characterize performance. Feature availability depends on the deployment, contract configuration, and finalized protocol state of the target network.

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1 Introduction

FrostBit combines an order book and settlement engine in its own chain. Transactions enter a bounded transaction pool. A proposer assembles a candidate block, and Snowman resolves which candidate extends the finalized chain. The virtual machine (VM) executes the ordered transactions, updates consensus state, and records results for queries. External assets enter and leave through a separate custody and verification workflow.

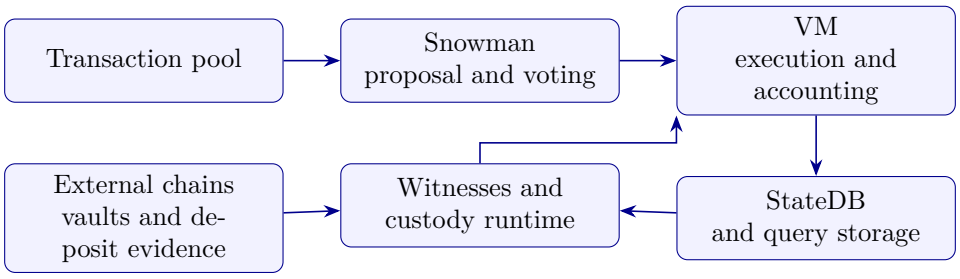


Figure 1: On-chain execution and external custody. External facts enter the VM through protocol transactions. The custody runtime reads finalized requests and produces authorization artifacts.

Three boundaries guide the design. A block must pass consensus and execution checks before changing the ledger. An external withdrawal requires authorizations from both the operator committee and the Guardian. A deposit credit depends on external evidence and the witness decision procedure. Each boundary has a different set of trust and availability assumptions. The tagged v0.8.4 source is the reference for the implementation described below.

2 Consensus Protocol

2.1 Candidate admission

FrostBit uses a chain-based implementation of the Snow family of protocols [1]. The live vote is for the next height above a finalized parent. A candidate is eligible only after validation, including a commitment to the ordered transaction body and consensus header and a comparison of the pre-execution state fingerprint against the local finalized state. The proposer eligibility check uses a verifiable random function (VRF) proof and a chain-registered public key. Proposal windows govern proposer eligibility. Voting on eligible candidates follows the decision rule below. Block commitment, candidate checks, and proposer validation specify the checks.

2.2 Poll sampling and vote authentication

An ordinary poll takes a snapshot of its parameters and samples K logical slots with replacement from the eligible peer set, excluding the requester. The requester sorts and deduplicates the source set, then mixes a seed tied to the poll context with a sequence number and a fresh random nonce. A peer hit multiple times receives one request. Its valid reply is counted with the slot weight registered by the requester.

The V2 query signatures bind the chain, request, both identities, and query context. The responder signs its Chit, and the requester accepts at most one valid reply per sampled identity in its current poll. Timeout or send failure leaves the affected slots as abstentions. The parameters remain fixed through settlement. Query handling and slot sampling implement this path.

2.3 Preference, confidence, and finalization

The default production profile caps the poll at $K = 15$, with a preference threshold of 8, a confidence threshold of 12, and a finalization threshold of $\beta = 15$ consecutive successful rounds. For a finalized parent with $N \geq 2$ Active and Probation voters, the default profile sets $K = \min(15, N-1, \lfloor (3N+2)/4 \rfloor)$. When $K < 15$, the preference threshold is $\lfloor K/2 \rfloor + 1$, and the confidence threshold is the larger of that threshold and $\lfloor (4K+2)/5 \rfloor$. Each poll records the resolved profile when it starts. Default configuration and profile resolution give the parameters.

The first candidate validated and registered initializes the preference with zero strength and confidence. A candidate reaching the preference threshold gains one unit of cumulative strength. A candidate replaces the current preference when its cumulative strength exceeds it. A candidate reaching the confidence threshold also advances the consecutive count for that candidate. A hard majority for a different candidate starts a new count. A round without a hard majority clears the consecutive count while preserving cumulative strength. The candidate with β consecutive hard-majority rounds is the one submitted for finalization. Finalization revalidates the block before execution and commit. The preference state machine defines these transitions.

2.4 Catch-up and checkpoint synchronization

A reported peer height serves as a lag hint. A node catching up by blocks starts an accepted-frontier round with a finalized local anchor, a target height, and a fresh nonce. This round samples distinct miners. The fixed confidence threshold must be met by distinct sampled identities signing support for the same target height and block ID. An authorized block-sync service peer can supply the block data. The node validates the downloaded blocks against the accepted target, their parents, and each prior state before applying them in height order. A quorum reporting a conflicting finalized anchor causes a safety stop that preserves local state. Each new batch uses a new round and anchor. Frontier synchronization defines the gate.

For deep lag, a node verifies a chain of checkpoint certificates and uses the certified state fingerprint to verify the downloaded StateDB snapshot. It then uses fresh accepted-frontier rounds for blocks after the checkpoint. Ordinary Chits retained after local finalization serve diagnostics. Synchronization uses the separate frontier and checkpoint protocols. See certificate verification, snapshot verification, and Chit persistence.

3 FROST Threshold Signatures

3.1 Operator and Guardian authorities

A vault operator key belongs to a committee of n miners. A qualifying set of at least t members can produce the operator signature using FROST [2]. Members hold individual key shares. The external vault also requires a distinct signature from the configured Guardian. Bitcoin uses a Taproot script with both checks. The account-chain vault contracts verify both signatures in contract code. Taproot construction and the account-chain vault contract show the authority checks.

Bitcoin BIP-340 uses x-only public keys, an even-y convention, and a tagged challenge hash [4]. The signing implementation must handle the corresponding key and nonce parity adjustments. The chain adapters and vault configuration distinguish the two signature families.

External network	Vault signature path
Bitcoin	BIP-340 Schnorr over secp256k1 in a Taproot script.
Ethereum, BNB Chain	alt_bn128 Schnorr in a vault contract.
Base, Arbitrum	alt_bn128 Schnorr in a vault contract.
TRON	alt_bn128 Schnorr for vault authorization. The outer TRON transaction uses secp256k1.

Table 1: Supported chain adapters and vault signature paths in the reference code. Actual use follows the target network's vault configuration.

3.2 Distributed key generation

A distributed key generation (DKG) ceremony uses secret sharing and commitments [5, 6]. In the polynomial construction, each dealer j samples a polynomial f_j of degree $t-1$ over the scalar field $\mathbb{Z}_q$. With group generator G, the dealer commits to its coefficients as $C_{j,k} = a_{j,k}G$ and delivers a share $s_{j\to i} = f_j(i)$ to recipient i. The recipient checks

$$s_{j\to i}G = \sum_{k=0}^{t-1} i^k C_{j,k}.$$

The accepted shares sum to the participant's share $s_i = \sum_j s_{j\to i}$, and the accepted constant commitments sum to the group public key $Q = \sum_j C_{j,0}$. The ceremony avoids assembling the group secret in a single process.

The chain records dealer commitments and recipient-encrypted shares. Recipients decrypt and verify their shares and the ceremony handles disputes. Public records expose commitments, outcomes, and the group public key. Plaintext share verification is performed by the recipient. DKG worker and dispute worker describe the implementation.

3.3 Two-round signing and ROAST coordination

Each signer i commits two fresh secret nonces with public commitments D_i and E_i. For a signing set S, a binding factor ρ_i commits the signer to the message and the commitment transcript. The group commitment is $R = \sum_{i\in S}(D_i + \rho_i E_i)$. Given the group challenge c and Lagrange coefficient λ_i, the signer contributes the scalar

$$z_i = d_i + \rho_i e_i + c\lambda_i s_i \pmod{q}.$$

For the basic construction, $z = \sum_{i\in S} z_i \bmod q$ satisfies $zG = R + cQ$. Chain-specific encodings and parity handling determine the wire signature. Nonces must be unique to a signing attempt.

ROAST coordinates signing attempts over available participants, allowing the coordinator to gather valid shares despite stalled or faulty participants [3]. Progress depends on reaching enough responsive honest signers and on communication. The independent Guardian remains necessary to complete the vault authorization. See the withdrawal worker and ROAST transport adapter.

4 Cross-Chain Withdrawals

4.1 Request, claim, and signed artifact

A user submits an on-chain withdrawal request identifying the external network, asset, amount, and recipient. The DEX debits the amount and queues the request. In the standard whole-request flow, a protocol claim reserves the selected withdrawal work and binds

it to a canonical transaction or batch template. The request enters `SIGNING`. The operator committee and Guardian authorize the same template. The VM validates the submitted result against the claim, template, vault authority, and chain-specific signature rules before recording `SIGNED`.

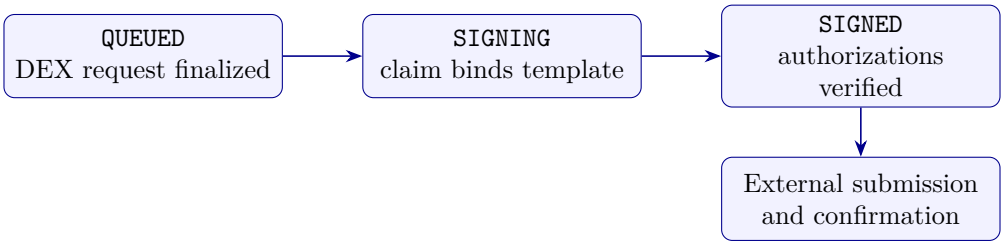


Figure 2: The standard whole-request withdrawal flow and its external execution step.

A signed record makes the execution artifact available. Delivery requires the external network to accept and execute the transaction. In this standard flow, the original requester may cancel while the request is queued and unclaimed. The order of the claim and cancellation transactions determines whether cancellation can take effect. Request, claim, signed record, and account-chain batch verification define these transitions.

4.2 Bitcoin

A BTC deposit address binds the DEX recipient to a vault-controlled Taproot output. The construction tweaks the operator key using the recipient identity and commits to a script that requires both the operator and Guardian signatures. On withdrawal, the planner selects eligible UTXOs and records their reservation. The FIFO path can combine multiple inputs to cover a batch. In the isolation path, single-input subtasks depend on the finalized protocol version and its activation receipt. A child task can be claimed while the parent remains queued. Cancellation in this mode refunds only the unallocated amount. The VM checks both signatures for each input. The signed transaction can be broadcast with its required dependencies, and confirmation is tracked on Bitcoin. The BTC miner fee is deducted from the withdrawal amount. Taproot scripts, BTC planning, isolation activation, and isolation cancellation specify these paths.

4.3 Account chains

On Ethereum, BNB Chain, Base, Arbitrum, and TRON, a signed batch is submitted to the configured vault contract. The contract checks the two authorizations over a domain-bound digest and enforces the applicable sequence and key epoch. V3 authority certificates advance epochs in sequence after the previous epoch's frozen batch cutoff is satisfied. A relayer or user's external-chain transaction executes the already-authorized transfer to the committed recipient. A DEX status or a submission hash can be reconciled with the contract's execution record on the destination chain. The V3 vault contract illustrates contract-side checks.

5 Deposit Witnesses

5.1 Evidence and selection

Bitcoin deposits are identified by their output, and account-chain deposits by the configured vault's deposit record. A DEX recharge request binds the external reference and intended recipient. Witnesses independently check the transaction or event, asset, amount, recipient, confirmations, and prior use of the reference. The VM applies the protocol's uniqueness rules

and credits balances through the finalized recharge path. Request validation and recharge finalization cover the state transitions.

The request fixes a future selection height. A beacon derived from finalized block hashes and the deposit reference feeds a reproducible ranking. For an original witness or ordinary arbitration committee, the ranking score is a hash of the beacon, request, round, and address multiplied by the candidate's registered stake. Address order breaks ties, and participants must be able to reserve the required case collateral. Using blocks finalized after the request reduces the requester's ability to search for favorable assignments in advance. The beacon's value as an entropy source depends on the underlying block production process. Beacon derivation and case selection give the details.

5.2 Voting and public challenge

Selected witnesses vote on the evidence. `PASS` means the deposit evidence is valid and `FAIL` means it is invalid. The protocol also supports abstention. An initial verdict enters a public challenge window. Challenge contributions are escrowed, and the required funding depends on the reserved bonds of the original committee. The v0.8.4 rules require at least 10 FB per new contribution and a funding threshold of the larger of 10 FB and 10% of the original committee's reserved case collateral. An underfunded challenge is refunded when its funding window ends. A deposit reaches final credit only after the case resolves to `PASS` and finalization updates balances. Protocol amounts, challenge funding, and final credit define these rules.

5.3 Ordinary arbitration and final court

Ordinary arbitration uses up to three rounds. The implementation defaults are planned committees of 4, 8, and 16 with a 75% same-direction threshold. The threshold uses the frozen planned seats. At the start of the third round, the four highest-staked eligible witnesses are frozen as the final court, with address order breaking stake ties. If the third ordinary round ends without a verdict, the case goes to this court. A final court verdict requires at least three members to agree in one ballot. An unresolved ballot advances the procedure without choosing a verdict. The frozen court may need another ballot, and insufficient availability can delay the outcome. Arbitration rules and round advancement define the escalation.

5.4 Settlement and case collateral

The witness fee is collected and escrowed before voting. At settlement it is divided equally among eligible participants whose votes agree with the final outcome. The VM tracks collateral reserved for each case. Wrong frozen positions can forfeit the relevant case bond. Silence or abstention alone carries no collateral penalty. An unchallenged case releases its assignments without wrong-vote slashing.

A successful challenge changes the original verdict and returns contributors' principal. The penalties actually collected are split among burn (10%), eligible contributors (45%, weighted by contribution), and eligible correct voters (45%, equal shares). When the challenge fails, its principal and collected penalties go to eligible correct voters. Pools with no eligible recipient are burned. Court members can earn an eligible witness fee and are excluded from penalty awards. These transfers are funded by escrow and actual collateral debits. Collateral accounting and reward allocation specify the eligibility and payout rules.

The launch-chain rule for chain ID 51025 adds a minimum 0.01 FB prepaid witness fee and a separate 10 FB requester bond. On final `PASS`, the bond is returned. On final `FAIL`, it is forfeited. On a successful recharge under this rule, the gross external amount funds separate one-basis-point awards for the requester and a pool shared by eligible participants

under the final `PASS` settlement rules, which can include ordinary arbitrators and final court members. Each award is rounded down to integer base units. The remainder is credited to the recipient. Launch-chain rule, requester bond, and recharge payout describe the scope and settlement.

6 Vault Key Management

6.1 Committee tenure

In rotating vault mode, the committee serves a term of 120,960 finalized blocks from actual activation. When the term is due, the protocol evaluates eligible mature candidates with locked stake. If the same group remains selected, it extends the review deadline and keeps the existing key and activation height. If there are too few eligible candidates, the current authority remains in place pending review. A qualifying membership change starts a new DKG and handover. The retained fixed-key genesis mode skips automatic rotation. Block-count constants and term processing define the policy.

6.2 Handover and miner exit

The new committee runs DKG for the next group key. The outgoing operator group and the configured Guardian authorize the handover. Finalization records the transition and activates the new authority on the DEX. External Bitcoin movements or contract epoch updates have their own execution and confirmation requirements. Existing external obligations can constrain which epoch can execute and when the handover can be applied there. Handover verification and contract sequencing show the separate checks.

Consensus liveness is tracked independently of vault tenure. Sufficient evidence of a miner's unavailability starts a 17,280-block grace period. After that period, sufficiently fresh offline evidence can lead to removal from active consensus participation. An accepted online report or a finalized block from the verified miner clears the offline start. The block counts depend on chain progress. An explicit miner exit is authorized by the hot miner identity and can be blocked by unresolved custody duties. The protocol checks the freeze and outstanding handovers before releasing remaining stake to the bound reward address. Offline grace and miner exit define the lifecycle.

7 On-Chain Trading and Asset Accounting

7.1 Spot and perpetual order books

The VM settles orders in the finalized transaction order. The matcher uses the best opposing price first, with deterministic order-index ordering for makers at the same price. The order index encodes price and order ID. An open order reserves funds. Fills update the order, the relevant balance buckets, and any remaining reservation. A partial fill leaves only the remaining quantity available to match.

Perpetual orders also reserve margin. A fill changes position size and margin and may realize profit or loss. Mark prices and maintenance margin inform liquidation. The VM uses a state view that includes earlier changes in the same block, so later orders and risk checks see the updated book and positions. Order index keys, matching, and the perpetual engine define these operations.

7.2 Funding, liquidation, and insurance

Funding periodically transfers value between the sides of an open perpetual market. The reference implementation defaults to a 10-block interval and a cap of 75 basis points per

settlement in either direction. The funding hook applies settlement to open positions in the affected market. Debits actually collected enter the market's insurance fund before payments to recipients are settled. When no payer can fund the distribution, cumulative funding can advance without creating a credit from nothing.

If a position falls below the maintenance requirement, liquidation reduces or closes the exposure. Margin written off by the liquidation goes to the insurance fund. A deficit beyond the bankruptcy price is tracked as a shortfall for the auto-deleveraging path. Funding parameters, block hooks, and funding and liquidation accounting define these rules.

7.3 Conservation and account lifecycle

For each token, the VM audits the block diff. It sums the changes in balance buckets, open-position margin, insurance balances, and active challenge escrow where applicable, and reconciles them with the change in total token supply. A nonzero net delta causes a hard execution failure. Fees flow through a runtime fee pool. Minting and burning require the corresponding supply changes, including cross-chain credits and debits that change the represented supply. Conservation audit and fee and execution accounting define this check.

State rent gives qualifying user accounts a height-based lifecycle. The protocol uses asset balances and applicable exemptions to determine which accounts owe rent. If an account reaches the cleanup stage, the protocol can cancel outstanding orders and transfer remaining assets to a protocol recovery account. Exemptions and the accounting of assets outside ordinary balances matter when determining whether an account is due for cleanup. Rent exemptions and rent processing define the account lifecycle.

8 Security Analysis

8.1 Consensus and synchronization assumptions

Repeated sampling gives probabilistic consensus behavior. The safety and progress analysis depends on the eligible voter set, adversarial participation, authenticated messages, sampling, and network delivery. A network disruption can delay successful polling and finalization. Nodes revalidate commitments, proposer eligibility, and prior state before applying a block. A sync node must validate an accepted-frontier round or a certificate-backed checkpoint as applicable. A conflicting frontier quorum at the local finalized anchor causes a safety stop. The sampling design and acceptance checks describe the implemented protocol. Its behavior under a deployment's fault and network model requires separate analysis.

8.2 Custody assumptions

The operator threshold distributes signing power among members. FROST security relies on the cryptographic assumptions of the signing scheme and on protecting shares and nonces [2]. The Guardian is a distinct trust and availability dependency. Compromise of the required authorizations can permit unauthorized movement, and unavailability of either required authority can delay withdrawals or handover. External scripts and contracts must enforce the intended authorization, replay protection, and execution rules for the configured network.

8.3 Deposit and settlement assumptions

Witness judgments depend on the accuracy and availability of external chain data, the confirmation policy, and the behavior of the original committee and arbitration bodies. Case bonds and the public challenge window create incentives and opportunities to correct bad outcomes. Arbitration may remain unresolved until enough of the assigned participants

submit consistent votes. Token conservation checks the DEX ledger's accounting transitions. External collateral and delivery must also be verified on their respective chains.

9 Performance Characteristics

The costs below describe the implemented work. They give protocol thresholds and scaling dimensions for measurement.

Path	**Cost driver or progress condition**
Ordinary polling	At most K query recipients and replies per requester per poll. Repeated logical slots share a network exchange. Gossip, transaction payloads, and retries have separate costs.
Block finalization	A candidate needs β consecutive hard-majority rounds. Proposal cadence, round timing, contention, and block execution all affect wall-clock delay.
Catch-up	Frontier rounds authorize target batches. Certificate and snapshot verification, download bytes, and replay determine deep-sync work.
Withdrawal	Planning, the number of inputs or outputs, signing attempts, Guardian availability, and external confirmation each contribute to end-to-end delay.
Trading and state	Work varies with transactions, matched orders, affected positions, and touched state. Consensus fingerprints cover live state, while order and trade history can use query storage.

Table 2: Implementation cost drivers. The table reports no wall-clock benchmark.

A comparable benchmark identifies the release and configuration, number and topology of voters, network delay and loss, transaction mix, offered and executed load, observation window, and hardware. For finalization, it reports the median and tail of time from a defined start event to local finalization. For withdrawals, it reports DEX signing and external confirmation separately. For deep sync, it reports starting lag, snapshot size, and elapsed time. The consensus simulator uses virtual time, so its timings describe the simulation.

10 Related Work

FrostBit's block voting draws on the Snow family [1]. Classical BFT protocols such as Tendermint and HotStuff offer useful points of comparison [7, 8]. Comparison requires matching network and adversary assumptions and distinguishing the ordinary polling path from synchronization.

The custody design builds on secret sharing, verifiable commitments, FROST signing, and ROAST coordination [2, 3, 5, 6]. FrostBit combines these building blocks with an independently authorized external vault, an on-chain handover policy, and a witness procedure for crediting deposits.

11 Conclusion

FrostBit combines sampled block finalization, deterministic exchange execution, and cross-chain custody. A user can follow a withdrawal from the DEX request through dual authorization to external execution. Deposit credits follow a separate evidence, challenge, and settlement process. The tagged source anchors the protocol rules in this edition. Network-specific configuration, activation records, and measurements are needed to evaluate a deployment.

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